STEFAN ZETZSCHE

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PROFILE

Applied Scientist in the Automated Reasoning Group at Amazon Web Services, working at the intersection of formal methods and AI. Currently developing Strata, a Lean-based intermediate language for verifying programs in mainstream languages; previously a core contributor to the Dafny verification language. Further work includes neuro-symbolic requirements analysis, verified algorithm development, and agentic property-based testing. Research published at ICML, NeurIPS, and PLDI.

EMPLOYMENT

Amazon Web Services, Applied Scientist — Dec 2022 - Present

- **Strata**: Develop an intermediate verification language written in Lean for verifying programs in mainstream languages (Java, Python, Rust). Strata models programs using composable intermediate representations, generates verification conditions, and discharges them via SMT solvers.
- **Dafny and Lean**: Core member of the Dafny development team, contributing to language design and verification infrastructure; organizer of the Dafny Workshop at POPL (2024–2026). Developed a verified probabilistic sampling library (VMC) in Dafny, later ported to Lean; the Lean version underpins verified foundations for differential privacy deployed in AWS Clean Rooms, receiving a Distinguished Artifact Award at PLDI 2025.
- **LLMs and Formal Methods**: Research spanning benchmarking LLM-generated formally verified code, LLM-guided theorem proving on competition mathematics, automated pipelines for large-scale verified code synthesis, and LLM pre-training on symbolic execution traces to improve code reasoning.
- **Automated Testing**: Apply agentic property-based testing to validate and axiomatise specifications and detect correctness and security issues at scale, uncovering multiple previously unknown Python standard library bugs.
- **Automated Reasoning Checks**: Contribute to a neuro-symbolic tool deployed in Amazon Bedrock that formally verifies LLM-generated responses against customer-defined policies, reducing hallucinations in production.
- **Kiro Requirements Analysis**: Design and implement neuro-symbolic requirements analysis deployed in Kiro (Amazon's agentic IDE): LLMs formalize natural-language specifications into SMT-lib, and an SMT solver detects inconsistencies and completeness gaps before code generation begins.

Meta, Software Engineer Intern — Jul 2022 - Sep 2022

- Enhanced a static analyzer for the Hack programming language, improving code reliability in Meta's core infrastructure.

Amazon Web Services, Applied Scientist Intern — Aug 2021 - Nov 2021

- Member of the Dafny core team, contributing to the language infrastructure, supervised by Rustan Leino.

VISITING POSITIONS

University College London, Visiting Researcher — Apr 2026 - Present

- Hosted by Prof. He Ye in the Software Systems Engineering Group.

EDUCATION

PhD Computer Science, University College London — 2018 - 2023

- Conducted research on automata learning, Kleene algebra, and category theory, culminating in the thesis *Canonical Algebraic Generators in Automata Learning*.
- Funded by VeTTs and ERC grants; member of the Programming Principles, Logic, and Verification Group. Advised by Alexandra Silva and Matteo Sammartino.

MSc Mathematics, University of Hamburg — 2016 - 2018

- Thesis on category theory (*Generalised Duality Theory for Monoidal Categories and Applications*), awarded top mark (1.0); graduated with highest distinction (overall grade 1.0).

BSc Mathematics, University of Hamburg — 2014 - 2016

- Thesis on group and graph theory (*Isomorphism Classes of Vertex-Transitive Tournaments*), awarded top mark (1.0).

PUBLICATIONS

Conferences

[C1] **MiniF2F-Dafny: LLM-Guided Mathematical Theorem Proving via Auto-Active Verification** 2026
M. Bakšys, S. Zetzsche, O. Bouissou, S. B. Holden
International Conference on Machine Learning (ICML)

[C2] **CLEVER: A Curated Benchmark for Formally Verified Code Generation** 2025
A. Thakur, J. Lee, G. Tsoukalas, M. Sistla, M. Zhao, S. Zetzsche, G. Durrett, Y. Yue, S. Chaudhuri
Conference on Neural Information Processing Systems (NeurIPS)

[C3] **Verified Foundations for Differential Privacy** 2025
M. Medeiros, M. Naveed, T. Lepoint, T. Kahsai, T. Ravitch, S. Zetzsche, A. Joshi, J. Tassarotti, A. Albarghouthi, J. Tristan
Distinguished Artifact Award
Conference on Programming Language Design and Implementation (PLDI)

[C4] **Compiler Fuzzing in Continuous Integration: a Case Study on Dafny** 2025
K. Boonriong, S. Zetzsche, A. F. Donaldson
International Conference on Software Testing, Verification and Validation (ICST)

[C5] **Well-Behaved (Co)algebraic Semantics of Regular Expressions in Dafny** 2024
S. Zetzsche, W. Różowski
International Colloquium on Theoretical Aspects of Computing (ICTAC)

[C6] **Generators and Bases for Monadic Closures** 2023
S. Zetzsche, A. Silva, M. Sammartino
Conference on Algebra and Coalgebra in Computer Science (CALCO)

[C7] **Guarded Kleene Algebra with Tests: Automata Learning** 2022
S. Zetzsche, A. Silva, M. Sammartino
Conference on Mathematical Foundations of Programming Semantics (MFPS)

[C8] **Canonical Automata via Distributive Law Homomorphisms** 2021
S. Zetzsche, G. v. Heerdt, M. Sammartino, A. Silva
Conference on Mathematical Foundations of Programming Semantics (MFPS)

Workshops

[W1] **Teaching LLMs Program Semantics via Symbolic Execution Traces** 2026
J. Bayer, S. Zetzsche, O. Bouissou, R. Delmas, M. Tautschnig, S. Kong
Deep Learning for Code Workshop at the International Conference on Machine Learning (ICML)

[W2] **ATLAS: Automated Toolkit for Large-Scale Verified Code Synthesis** 2026
M. Bakšys, S. Zetzsche, O. Bouissou, S. Kong, R. Delmas
- Deep Learning for Code Workshop at the International Conference on Machine Learning (ICML)
- Dafny Workshop at the Symposium on Principles of Programming Languages (POPL)

[W3] **MiniF2F-Dafny: LLM-Guided Mathematical Theorem Proving via Auto-Active Verification** 2026
M. Bakšys, S. Zetzsche, O. Bouissou
Dafny Workshop at the Symposium on Principles of Programming Languages (POPL)

[W4] **DafnyPro: LLM-Assisted Automated Verification for Dafny Programs** 2026
D. Banerjee, O. Bouissou, S. Zetzsche
Dafny Workshop at the Symposium on Principles of Programming Languages (POPL)

[W5] **CLEVER: A Curated Benchmark for Formally Verified Code Generation** 2025
A. Thakur, J. Lee, G. Tsoukalas, M. Sistla, M. Zhao, S. Zetzsche, G. Durrett, Y. Yue, S. Chaudhuri
AI for Math Workshop at the International Conference on Machine Learning (ICML)

[W6] **Well-Behaved (Co)algebraic Semantics of Regular Expressions in Dafny** 2025
S. Zetzsche, W. Różowski
Dafny Workshop at the Symposium on Principles of Programming Languages (POPL)

[W7] **Verifying the Fisher-Yates Shuffle Algorithm in Dafny** 2025
S. Zetzsche, T. Lepoint, J. Tristan, M. Mayer
Dafny Workshop at the Symposium on Principles of Programming Languages (POPL)

[W8] **Dafny as Verification-Aware Intermediate Language for Code Generation** 2025
Y. C. Li, S. Zetzsche, S. Somayyajula
Dafny Workshop at the Symposium on Principles of Programming Languages (POPL)

[W9] **Randomised Testing of the Dafny Compiler: Into the CI** 2025
K. Boonriong, A. F. Donaldson, S. Zetzsche
Dafny Workshop at the Symposium on Principles of Programming Languages (POPL)

[W10] **VMC: a Dafny Library for Verified Monte Carlo Algorithms** 2024
F. Zaiser, S. Zetzsche, J. Tristan
Dafny Workshop at the Symposium on Principles of Programming Languages (POPL)

SUPERVISION

Interns at Amazon Web Services

- Jonas Bayer (PhD Student at the University of Cambridge) 2025
- Mantas Bakšys (PhD Student at the University of Cambridge) 2025
- Debangshu Banerjee (PhD Student at the University of Illinois Urbana-Champaign) 2025
- Yue Chen Li (Undergraduate Student at MIT) 2024
- Wojciech Różowski (PhD Student at the University College London) 2024
- Fabian Zaiser (PhD Student at the University of Oxford) 2023
- Yann Herklotz (PhD Student at the Imperial College London) 2022

PROFESSIONAL SERVICES

Program Committee Chair

- Dafny Workshop at the Symposium on Principles of Programming Languages (POPL) 2026
- Dafny Workshop at the Symposium on Principles of Programming Languages (POPL) 2025
- Dafny Workshop at the Symposium on Principles of Programming Languages (POPL) 2024

Program and (Sub-)Review Committee Member

- International Conference on Formal Methods in Software Engineering (FormaliSE) 2027
- International Conference on Certified Programs and Proofs (CPP) 2027
- International Conference on Verified Software: Theories, Tools and Experiments (VSTTE) 2026
- International Workshop on Language Models and Programming Languages (LMPL) 2026
- Annual Conference on Neural Information Processing Systems (NeurIPS) 2026
- AI for Math Workshop at the International Conference on Machine Learning (ICML) 2026
- Deep Learning for Code Workshop at the International Conference on Machine Learning (ICML) 2026
- AI for Math Workshop at the International Conference on Machine Learning (ICML) 2025
- International Conference on Computer-Aided Verification (CAV) 2025
- Conference on Algebra and Coalgebra in Computer Science (CALCO) 2023
- International Symposium on Model Checking of Software (SPIN) 2023

Mentor

- Symposium on Principles of Programming Languages (POPL) 2021
- Conference on Systems, Programming, Languages and Applications: Software for Humanity (SPLASH) 2020

Organizer

- Dafny Workshop at the Symposium on Principles of Programming Languages (POPL) 2026
- South of England Regional Programming Languages Seminar (SREPLS) 2025
- Dafny Workshop at the Symposium on Principles of Programming Languages (POPL) 2025

- Dafny Workshop at the Symposium on Principles of Programming Languages (POPL) 2024

Artifact Evaluator

- International Conference on Computer-Aided Verification (CAV) 2025
- Conference on Programming Language Design and Implementation (PLDI) 2025
- International Conference on Functional Programming (ICFP) 2025
- Symposium on Principles of Programming Languages (POPL) 2025
- Conference on Systems, Programming, Languages and Applications: Software for Humanity (SPLASH) 2025
- International Conference on Functional Programming (ICFP) 2024
- Conference on Programming Language Design and Implementation (PLDI) 2024
- Symposium on Principles of Programming Languages (POPL) 2024
- International Conference on Computer-Aided Verification (CAV) 2023
- International Conference on Computer-Aided Verification (CAV) 2022

TEACHING

Teaching Assistant at University College London

- Logic and Database Theory 2020
- Discrete Mathematics for Computer Scientists 2020
- Computability and Complexity 2020
- Theory of Computation 2019
- Principles of Programming 2019
- Discrete Mathematics for Computer Scientists 2018

Teaching Assistant at University of Hamburg

- Analysis I 2017
- Linear Algebra and Analytic Geometry II 2016
- Linear Algebra and Analytic Geometry I 2015